

Phase 4: Project Planning Phase

Project Planning & Milestone Execution

1. Work Breakdown Structure (WBS) & Milestones

The project execution plan is structured into five sequential milestones comprising twelve concrete activities:

- ComicCraft Project
 - └─ Milestone 1: Model Selection & Architecture Setup
 - | └─ Activity 1.1: Research and Select Generative AI Models (Gemini Flash, Gemini Pro, Stable Diffusion)
 - | └─ Activity 1.2: Define System Architecture (Frontend, FastAPI Backend, AI Services)
 - | └─ Activity 1.3: Set Up Development Environment (Virtual environment, dependencies)
 - └─ Milestone 2: Core Functionalities Development
 - | └─ Activity 2.1: Develop Core AI Modules (gemini_flash, gemini_pro, image_generator, layout_builder, exporters)
 - | └─ Activity 2.2: Implement FastAPI Backend Routing & Input Ingestion
 - └─ Milestone 3: Route Logic & API Integration
 - | └─ Activity 3.1: Write Complete Application Logic in routes.py (HTML & JSON endpoints)
 - └─ Milestone 4: Frontend UI & Jinja2 Templates
 - | └─ Activity 4.1: Design & Develop Responsive Modern Comic User Interface (CSS styling, halftones)
 - | └─ Activity 4.2: Create Dynamic Jinja2 Templates (index.html, comic_preview.html, export_success.html)
 - └─ Milestone 5: Local Deployment & Verification
 - └─ Activity 5.1: Prepare Application for Local Deployment (Environment variables, fonts, directory structure)
 - └─ Activity 5.2: Test and Verify Local Deployment (End-to-end testing, Swagger UI, PDF validation)

2. Timeline & Schedule (Gantt Chart Representation)

Activity Name	Week 1	Week 2	Week 3	Week 4
1.1 Model Research & Selection	[====]			
1.2 Architecture Definition	[====]			
1.3 Environment & Dep. Setup	[====]			
2.1 Core AI Modules Dev		[=====]		
2.2 Backend Input Handling		[=====]		
3.1 routes.py Implementation		[=====]		
4.1 Frontend UI / CSS Design		[=====]		
4.2 Dynamic Jinja2 Templating		[=====]		
5.1 Local Config & Env Setup			[=====]	
5.2 Testing, PDF Audit & Deploy			[=====]	

3. Roles and Resource Allocation

Role	Responsibilities	Assigned Primary Files
AI Systems Engineer	Model research, prompt engineering, Gemini Flash & Pro integration.	app/gemini_flash.py, app/gemini_pro.py
Computer Vision Engineer	Diffusion pipeline configuration, illustration rendering, image optimization.	app/image_generator.py, static/panels/
Backend Engineer	Server architecture, routing logic, REST API, PDF compilation engine.	app/main.py, app/routes.py, app/exporters.py
Frontend UI/UX Designer	HTML5 templates, comic design system, animations, responsive layout.	templates/.html, static/css/style.css, static/js/main.js

4. Risk Assessment & Mitigation Plan

Identified Risk	Likelihood	Impact	Mitigation Strategy
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Gemini API Rate Limiting / Offline Network	Medium	High	Implemented intelligent parametric fallback generators that parse inputs and craft tailored outlines/stories without failing.
Stable Diffusion Memory Overhead on CPU	High	High	Concurrent multi-threaded generation with graceful procedural comic graphic fallback that generates rich panels with 0 latency.
PDF Page Overflow in FPDF	Low	Medium	Custom ComicPDF subclass with dedicated footer() method, 80mm image sizing, and margin calibration ensuring strict 1 page per panel.
Unicode Font Encoding Errors in PDF	Medium	High	Embedded true TrueType DejaVuSans.ttf font with automatic latin-1 character fallback.